



Ste Marthe September 5, 2023

**RESEARCH PROJECT COMPARISON OF VITAL HUMIC_{tm} HYDRO
BIO
AND
Real Growers Recharge Natural Plant Growth Stimulant
AKA G2 PRODUCT**

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RESEARCH PROJECT FOCUS:

PRODUCTIVITY, COST, OVERALL PERFORMANCE

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HydroBio-Vital Humic, G2 Competitor Comparison

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OVERVIEW

Compare, contrast, and analyze the key differences and potential benefits of using nutrient supplementation product HydroBio, by Vital Humic against a similar and competitive product within the commercial cannabis cultivation industry, specifically Recharge by Real Growers; hereinafter referred to as Competitor. A side-by-side comparison study for propagation, cultivation and extraction was conducted using a single area, separated into three distinct areas, one control group, Test Group 1 using HydroBio with proprietary nutrient protocols and Test Group 2 using a competitive additive to HydroBio with the same proprietary protocol.

GOALS

1. Determine overall costs associated with using HydroBio vs competitive products.
2. Provide insight to increased plant vigor and overall growth.
3. Assess resin gland production, specific to capitate stalked trichomes through water extraction.
4. Analyze overall potency of finished cannabis flower and extract.

SPECIFICATIONS

Environment

Each cultivation environment consisted of a 500 square foot flowering space, utilizing 2 square feet per flowering plant: allowing a total of 250 flowering plants per designated area. Coco coir with 30% perlite was used as the cultivation medium with a seven-gallon pot, centralized drip irrigation, 20 x Pinnacle HP 600w LED Horticulture Lights. Temperature and humidity were maintained and controlled using industry standard specifications consistent with the plants' stage of growth.

Propagation

All propagation was conducted in a centralized environment, using manual fertigation/irrigation methods. All three groups used an aeroponic rooting system, Clonex Rooting Gel and Clonex Cloning Nutrients for the propagation process. Test Group 1 utilized .15g of HydroBio per gallon of fertilizer. Test Group 2 utilized the competitive nutrient additive at the recommended dosage of 3g / ½tsp per gallon.

Table 1. Input Cost Comparison Propagation 12-gallon Reservoir

	Cost per Gallon Mixed	Cost per Weekly Cycle	Yearly Cost
HydroBio	\$.02	\$1.80	\$93.60
Compet.	\$.27	\$3.24	\$168.48

Vegetative Growth

Vegetative growth was conducted in individualized areas specified above. The nutrient protocols utilized a proprietary blend of N,P,K in appropriate ratios for the vegetative phase of growth. In addition to the proprietary blend, Test Group 1 used HydroBio with every feeding/watering at .15g per gallon. Test Group 2 used a competitor product at the recommended dose of 6g / 1tsp per gallon, per week. All groups used a 300-gallon reservoir, replenished weekly as per standard operating procedures.

Table 2. Input Cost Comparison Vegetative Growth; 250 Cannabis Plants

	Cost per Gallon Mixed	Cost per Weekly Cycle	Yearly Cost
HydroBio	\$.02	\$2.50	\$130.0
Compet.	\$.54	\$67.50	\$3,510

*.5 gallon per plant, per day

Bloom Growth

Bloom growth was conducted in individualized areas specified above. The nutrient protocols utilized a proprietary

blend of N,P,K in appropriate ratios for the bloom phase of growth. In addition to the proprietary blend, Test Group 1 used HydroBio with every feeding/watering at .3g per gallon. Test Group 2 used a competitor product at the recommended dose of 6g/1tsp per gallon, per week. Competitor product was removed from fertigation protocol during the final two weeks of the bloom cycle, as per the manufacturer recommendations. All groups used a dedicated 300-gallon reservoir, replenished weekly as per standard operating procedures.

Table 3. Input Cost Comparison Bloom Growth; 250 Cannabis Plants

	Cost per Gallon Mixed	Cost per Weekly Cycle	Yearly Cost
HydroBio	\$.02	\$5.00	\$260.0
Compet.	\$.54	\$135.0	\$7,020

*1 gallon per plant, per day

Integrated Pest Management

Insects, pests, molds and fungi management protocols utilized a revolving weekly schedule of Neem Oil, Horticultural Oil, Diatomaceous Earth, Thieves Oil, Peppermint Oil, Rosemary Oil, Lady Bugs and Green Lace Wigs.

Genetics

In order to ensure adequate and comparable results, a single chemovar was selected for all groups. Each environment consisted of 250x Bergy's Gift (Sunset

Sherbert x (GG#4 x Chem's Sis)) cannabis plants.

Extraction

Each group's finished flower was divided in half to undergo two separate extraction methods: Fresh Frozen Bubble Hash and Liquid CO2 extraction. This was done to measure potencies, resin production/yield and cannabinoid recovery efficiencies.

Results

Control Group

The control group was initiated in an effort to establish baseline data to ensure the variable being studied is not changed or limited to zero. Further, to ensure the variables can be studied appropriately.

Control Group propagation provided insight and baselines for plant vigor, root growth and overall plant health. This standard method proved to provide quality, rooted clones, ready for transplant in 11 days on average.



Proprietary fertigation protocols provided a standard N,P,K delivery during the vegetative growth process. Fertigation baseline parameters were set with a target of 6.4pH, 2.0 EC and 660 ppm. Vegetative growth was maintained for 28 days with a PPFD of 600.

The Control Group's bloom cycle was maintained for 63 days with a PPFD target of 875. Again, proprietary fertigation protocols provided a baseline of 6.4pH, 3.1 EC and 975 PPM.



Once harvested, the Control Group's large fan leaves were removed, the flower was separated from the branches and weighed; 54,943g. Fifty percent (27,471g) of the wet material was used to produce Fresh Frozen Bubble Hash for resin volume analysis, and 50% was placed into a freeze dryer for pre-extraction processing. Liquid CO2 extraction was used in an effort to establish an extract baseline potency. The freeze-dried material produced a homogenized potency of 24% Total Active Cannabinoids (22.4% THCA, 1.7% THCD9, .1% CBGA). Further resulting in 3,840g of CO2 extract with a potency of 72% TAC. The Fresh Frozen Bubble Hash yielded 992g of finished product with 61% TAC.

Test Group 1

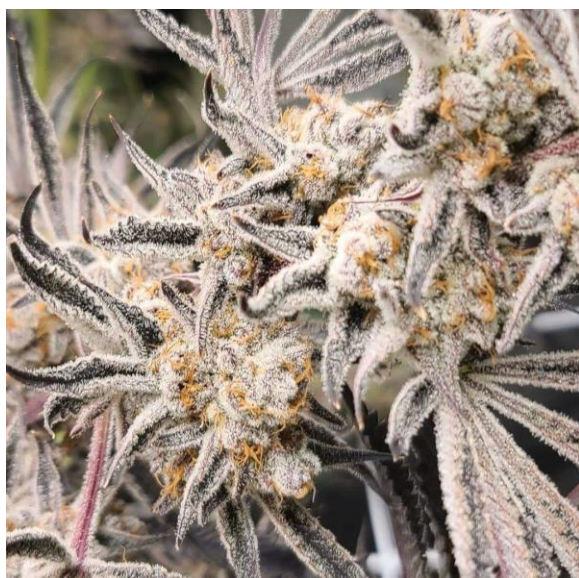
Test Group 1's propagation utilized the same method used in the Control Group with the addition of HydroBio. This method displayed rapid root development and visual plant health, with no signs of pests or fungal issues. Adding .15g of HydroBio per gallon, proved to provide quality, rooted clones, ready for transplant in 6 days.



Initial fertigation baseline parameters were set using the Control Group's recipe: 6.4pH,

2.0 EC and 660 ppm. In addition to proprietary fertigation protocols, adding HydroBio at a rate of .3g per gallon provided visually rapid growth and vigor not seen within the Control Group. Vegetative growth was maintained for 28 days. Each week, the PPM and EC were slightly increased in an effort to force a nutrient lockout scenario, ultimately reaching 1400 PPMs and 3.1 EC, with no observable nutrient lockout. The rapid and sustained uptake of nutrients appeared to also increase the plant's capacity to handle a more intense PPFD. Like the nutrient recipe, Test Group 1's PAR was baselined with the Control Group protocol, ultimately reaching a PPFD of 1050 in the final week of vegetative growth with no signs of reduced growth, vigor, or overall health.

Bloom cycle continued to show rapid development and true genetic chemovar potential. HydroBio allowed the plants to maximize all available nutrients and process a full spectrum of high intensity LED lighting. Ultimately reaching a PPM of 1250, and PPFD of 1200, the plants continued to show maximum potential and perfect health.



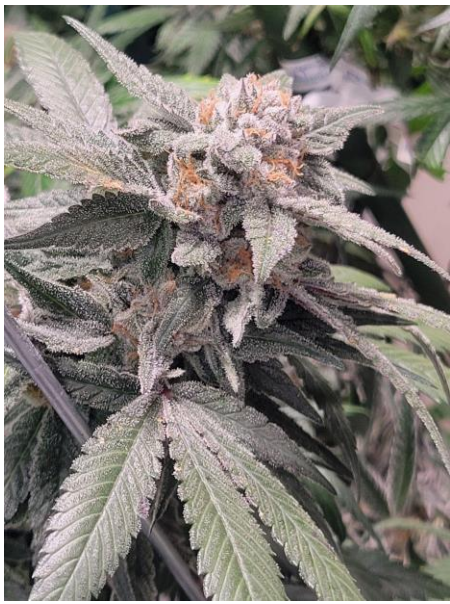
Once harvested, Test Group 1's large fan leaves were removed, the flower was separated from the branches and weighed; 68,130g. Fifty percent (34,650g) of the wet material was used to produce Fresh Frozen Bubble Hash for resin volume analysis, and 50% was placed into a freeze dryer for pre-extraction processing. Liquid CO2 extraction was used in an effort to compare against the Control Group's established baseline for extract potency. The freeze-dried material produced a homogenized potency of 32% Total Active Cannabinoids (29.4% THCA, 2.5% THCD9, .1% CBG). Further resulting in 5753g of CO2 extract with a potency of 78% TAC. The Fresh Frozen Bubble Hash yielded 2838g of finished product with 66% TAC.

Test Group 2

Test Group 2's propagation showed plant vigor, root growth and overall plant health. This standard method, along with adding .5 tsp of competitive additive per gallon, provided rooted clones, ready for transplant in 9 days.



Initial fertigation baseline parameters were set using the Control Group's recipe: 6.4pH, 2.0 EC and 660 ppm. In addition to proprietary fertigation protocols, adding the competitive additive at a rate of .5 tsp per gallon. Although the growth appeared to be faster and more vigorous than the Control Group, Test Group 1 appeared a week or two ahead. Vegetative growth was maintained for 28 days. Like Test Group 1, each week, the PPM and EC were slightly increased in an effort to force a nutrient lockout scenario, ultimately reaching 1150 PPMs and 3.2 EC, before signs of nutrient lockout became visible. The plants were flushed with pH'd water only to remove excess nutrient build-up and the nutrient regimen was continued, maintaining 1000 PPMs and 2.8 EC for the duration of the cycle. The nutrient lockout and subsequent flush also required the lights to maintain no more than 1000 PPFD.



Once harvested, Test Group 1's large fan leaves were removed, the flower was separated from the branches and weighed; 61,812g. Fifty percent (30,906g) of the wet material was used to produce Fresh Frozen Bubble Hash for resin volume analysis, and 50% was placed into a freeze dryer for pre-extraction processing. Liquid CO2 extraction was used to compare potency against the Control Groups established baseline. The freeze-dried material produced a homogenized potency of 28% Total Active Cannabinoids (26.4% THCA, 1.4% THCD9, .2% CBG). Further resulting in 4,567g of CO2 extract with a potency of 74% TAC. The Fresh Frozen Bubble Hash yielded 1,545g of finished product with 63% TAC.

Conclusion

Utilization of inputs beyond the standard N,P,K protocols has become standard practice within the agricultural industry, to include cannabis cultivation. As the cannabis crop continues to become more of a commodity, identifying opportunities to

reduce overhead and operational expenditures becomes more necessary. When considering the cost of equipment, labor, electricity, fertigation, irrigation, processing and packaging, limitations on cost reduction opportunities become more apparent. As international supply chain costs continue to increase, identifying sustainable alternatives and cost-effective nutrient options becomes paramount to a company's ability to maintain not only the quality of their products but also forward progression and continued growth.

Producing dried cannabis flower at sub \$0.50 per dried gram has become the only way to effectively meet projections and allow for competitive pricing and company growth. This price point makes the market itself extremely competitive, further requiring additives like Hydro Bio by Vital humic to help ensure targets are met.

When comparing HydroBio with other top market competitors, it is important to not only consider the nutrient dosing costs, on a per gallon basis, but also potential improvements in quality of outputs/products. A higher quality output can play a vital role in increased margin opportunities. With this study, we compared dosing costs on a per gallon basis, overall weight yields, potency yields and resin production.

Table 4. Annual Input Cost Comparison (Propagation, Vegetative and Bloom)

	Cost Per Gal	Weekly Cost	Annual Cost
Control	N/A	N/A	N/A
Test 1	\$.02	\$9.30	\$483.60
Test 2	\$.54	\$205.74	\$10,698

* All Growth Stage Requirements for 500 sq ft of canopy

With current cannabis trends, identifying chemovars and methods to achieve top quality flower with high cannabinoid and terpene potencies is essential. Although chemovars play an important role in trichome gland development, a targeted nutrient and supplement program can help enhance the plants' natural abilities. Delivering the proper microbes, humates, and carbohydrates at the right time will help deliver the chemovars full and true cannabinoid and terpene expressions.

Table 5. Flower Comparison (Cannabinoid Potency (TAC), Terpene Potency)

	TAC	Terpene Content
Control	24%	3.2%
Test 1	32%	6.2%
Test 2	28%	4.7%

*TAC - Total Active Cannabinoids

CO2 extraction yields indicate the trichomes were resin filled and abundant. Typically, CO2 is used for large, fast production with the intent of distillate manufacturing. However, CO2 can also produce high quality concentrates and is extremely efficient in extracting all available cannabinoid and terpene content within the plant matrix. The comparison in Table 6 provides insight into the effectiveness of HydroBio when used throughout the entire plant lifecycle.

Table 6. CO2 Extraction Comparison

	CO2 TAC	Yield %	Yield
Control	72%	22%	3,480g
Test 1	78%	30%	5,753g
Test 2	74%	26%	4,567g

* CO2 extraction efficiency of 95%

Water extraction was used with the intent of identifying the difference in raw resin production and recovery potential. Because this method simply separates trichome heads from the plant material, it is an ideal way to see the potential variations between product capabilities when seeking to increase overall trichome production.

Table 6. Water Extraction Comparison

	Water TAC	CO2 Terp	Yield
Control	72%	6.5%	992g
Test 1	78%	15%	2,838g
Test 2	74%	9%	1,545g

* Average Fresh Frozen return is 5%



COMMENTS

From *Mr. Patrice Kieffer VP & CEO*,

It is of most importance in this new economy of medical and recreational Cannabis in Canada and the USA that costing, effectiveness and overall performance of a Bio Stimulant be evaluated for the purpose of proving its overall performance versus other products offered to the small, medium, and large scale producers in the "cannabis industry". The open market has become more price competitive to the end user and cost of production are on the rise in Canada and the USA.

WE at ALC BIO INNOVATIONS are conscience that for a small, medium, and even a large public growing facility to be successful/profitable, the operations have to be able to produce a 1 gr of dried flower for 0,50 Cents USD or under.

This particular research project clearly shows all of the overall advantages, not only in productivity but also fiscally gained by utilizing VITAL HUMICTM HYDRO BIO versus a known competitor's bio stimulant.

Numbers speak louder than words...

Sincerely yours

Patrice Kieffer

VP & CEO

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